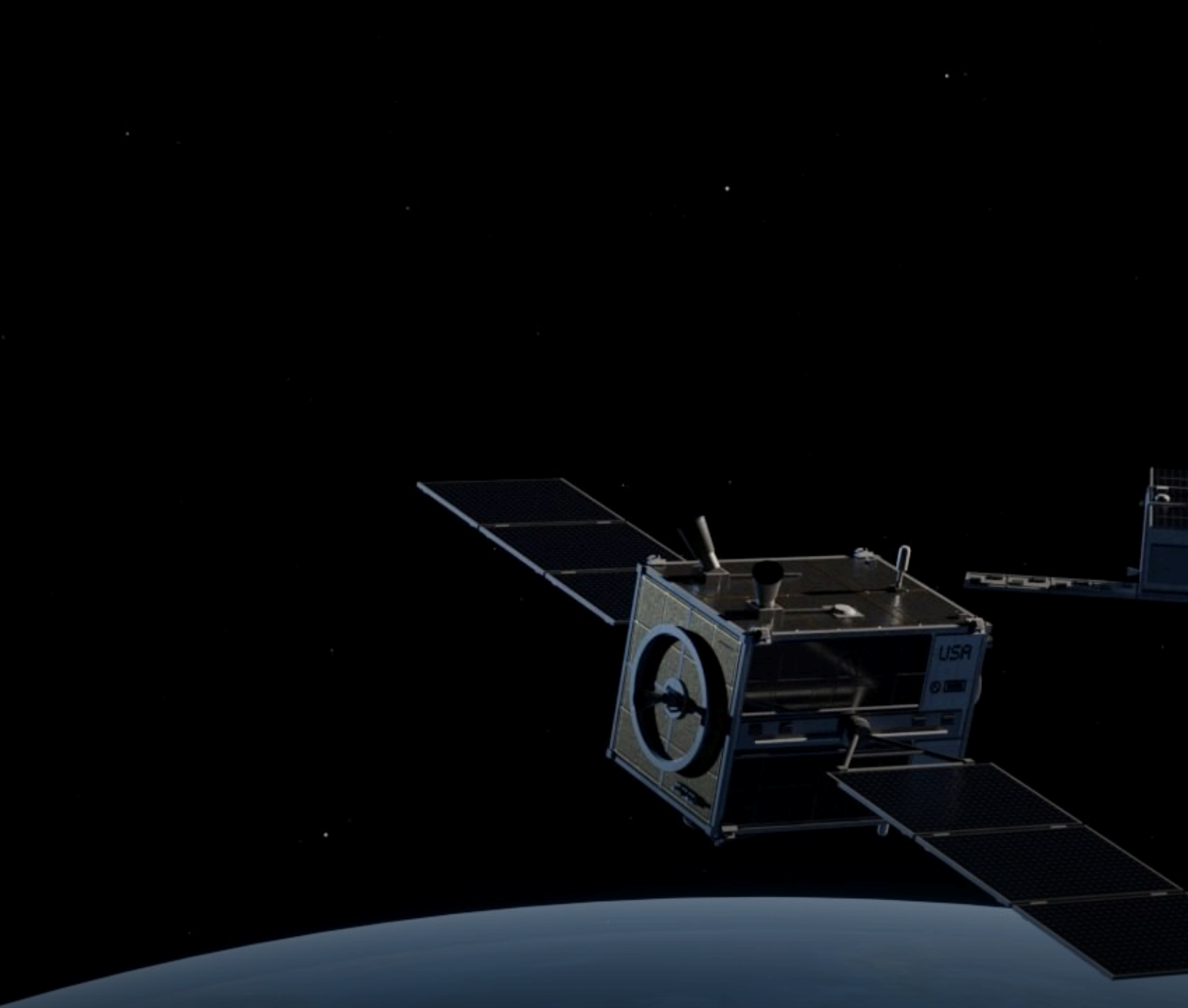


OPERATOR'S HANDBOOK

xGEO

LIVE

How to play CLOSE PASS: shadow a Russian inspector satellite in low Earth orbit and make it leave, with no weapons and without starting a war.

ALPHA 0.5 · BUILD 11 · KEYBOARD & MOUSE

WELCOME

BEFORE YOU PLAY

You fly one US inspector satellite. Another nation's inspector is creeping up on a satellite you protect. Your job is to make it leave, with no weapons and without starting a war.

WHAT KIND OF GAME THIS IS

There are no lasers and no explosions. Satellites race round the Earth at nearly 8 km/s, but they meet and part slowly, a few metres a second at a time.

Every move costs fuel you can never get back. Every look at the enemy depends on where the Sun is. Every radio burst you hear is worked out from real signal physics.

You don't need to know orbital mechanics. The game does the maths and tells you the price. This book gives you the few rules of thumb that make you good at it.

HOW TO READ THIS BOOK

1. **READ PAGES 4 TO 9**
   The mission, how you win, and how your opponent thinks. Ten minutes.
2. **PLAY WITH THE TUTORIAL ON**
   A new game teaches the controls in 24 short steps, one at a time.
3. **FOLLOW THE FIRST GAME**
   Part 2 walks you through a winning game, move by move.
4. **KEEP THE REST FOR LATER**
   Part 3 explains each tool. Part 4 is the cheat sheet, tips and glossary.

WHAT YOU NEED

A COMPUTER
Desktop or laptop. Phones and touch screens aren't supported.

A BROWSER
Current Chrome, Edge or Firefox. The first load takes about a minute.

MOUSE & KEYS
A scroll wheel for zooming. A middle button helps.

A BIG SCREEN
1920 × 1080 or more is best. F11 for full screen.

FIRST CLICK
Click inside the game once after it loads so it gets your keyboard. Until then some browsers keep F1 to F7 for themselves.

CONTENTS

WHAT'S INSIDE

Labels you see on screen are written LIKE THIS. Keys look like F3.

PART 1 · THE MISSION

The situation 5
How you win 6
Your opponent 7
Tension and war 8
Three ways to play 9

PART 2 · YOUR FIRST GAME

Starting a game 11
The screen 12
Time and the camera 13
Step by step 1: get your bearings 14
Step by step 2: close in 15
Step by step 3: see it off 16

PART 3 · YOUR TOOLS

Know where it is 18
Plan a move 19
Check the plan, then burn 20
Small moves up close 21
Look with the telescope 22
Listen to its radio 23
Identify the signal 24
Being seen 25
Radars and your track 26
The timeline 27
Other windows 28

PART 4 · REFERENCE

Playing well 30
The debrief 31
Physics you'll feel 32
Problems and answers 33
Cheat sheet 34
Glossary 35
The tutorial at a glance 36
For tinkerers 37



1

THE MISSION

Who's out there, what they want, and what counts as winning.

5 THE SITUATION

6 HOW YOU WIN

7 YOUR OPPONENT

8 TENSION AND WAR

9 THREE WAYS TO PLAY

1 · THE MISSION

THE SITUATION

Monday 26 October 2026, 21:00 UTC, about 540 km up. A Russian inspector is working its way round a US radar satellite. You have two days to make it leave.

This much is real. In October 2026 the Russian satellite Cosmos 2612 was reported closing on the commercial radar satellite ICEYE-X36.

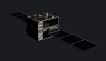
The rest is the game. The US sends its own inspector, WATCHMAN-1, to stand beside X36 and make the visitor's job hard.

Cosmos 2612 wants three close looks at X36. It doesn't want company. Get close enough, often enough, and it gives up and goes home.

But every time you crowd it, tension rises in both capitals. Push too hard and it ends in open conflict, which loses the game outright.

> **YOUR ORDERS**
> Make Cosmos 2612 break off before it finishes its look at ICEYE-X36. You have no weapons. No war.

WHO'S WHO



YOU · WCH-1

WATCHMAN-1

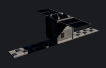
A US inspector (invented). About 260 m/s of fuel for the whole mission, a telescope and a radio receiver. No weapons, no radar.



THE VISITOR · TRK 4423

COSMOS 2612

A real Russian inspector. Only about 90 m/s of fuel. Watches you with a small infrared telescope and radios home on 2,250 MHz.



PROTECT · X36

ICEYE-X36

A real commercial radar satellite. It has no engine, so it can't get out of the way. You must keep the visitor off it.



YOUR EYES · SITES

FIVE GROUND RADARS

Real sites in Alaska, Texas, Arizona, New Zealand and the Pacific. Each sees Cosmos only while it passes overhead.

1 · THE MISSION

HOW YOU WIN

Five goals, 100 points. Two of them end the game if you fail. Press F1 during play to see them ticked off.

40 points

MAKE IT LEAVE MUST NOT FAIL

Lost if it finishes all three of its looks. Half marks if it's still at work when the two days end.

20 points

KEEP IT MORE THAN 1 KM FROM X36

Full marks if it never gets inside 1 km. At 500 m you keep only a quarter.

20 points

KEEP A GOOD TRACK ON IT

Know where it is to within 1 km at least 80 % of the time.

10 points

SAVE FUEL

Full marks using a quarter of your fuel or less. Fails past half.

10 points

NO WAR MUST NOT FAIL

Lost if escalation reaches 100. The lower it ends, the more you keep.

THREE WAYS IT ENDS

OPERATION COMPLETE
Cosmos 2612 gave up and left. You're scored on all five goals.

OPERATION FAILED
It finished its look, or escalation hit 100. Either ends the game at once.

THE 2 DAYS ARE UP
Neither happened in time. Half the points for the first goal.

A GOOD SCORE

It leaves (40), never gets inside 1 km of X36 (20), a good track 76 % of the time (19), 30 % of your fuel used (9), escalation ends at 15 (8.5). **Total 97.**

1 · THE MISSION

YOUR OPPONENT

Cosmos 2612 follows a few simple rules. Learn them and you know most of the game.

ITS PLAN: THREE LOOKS AT X36

STOP 1
2 KM BEHIND
Holds 6 hours

›

STOP 2
2 KM AHEAD
Holds 6 hours

›

STOP 3
0.8 KM BEHIND
Holds 8 hours. Inside X36's 1 km keep-out.

›

THEN
IT GOES HOME
About 29 hours in. You lose.

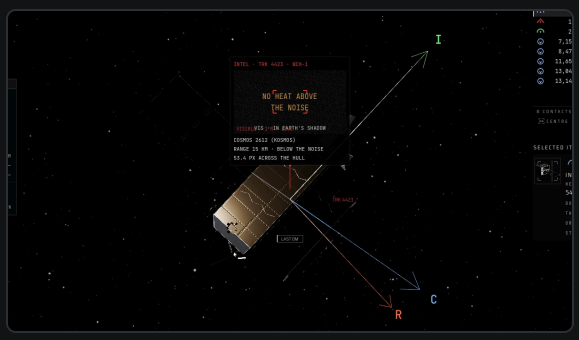
Each hop between stops takes about 3 hours. **Only a parked Cosmos can be pushed away**, so watch for the log line saying it has parked.

WHEN YOU CROWD IT

1ST TIME
IT GIVES WAY
It moves 8 km off, to the far side of X36. That takes about 2 hours, then it holds there 4 hours.

2ND TIME
IT GIVES WAY AGAIN
Same again: a new spot 8 km off on the far side.

3RD TIME
IT GIVES UP
A 4 m/s burn and it climbs away. 200 km from X36, it's gone and you win.



Cosmos 2612 seen from WATCHMAN-1, 15 km out.

WHAT COUNTS AS CROWDING

All three at once: it is **parked**, you are **within 5 km** of it, and it **can see you**. Mid-hop it ignores you.

WHEN IT'S LEFT ALONE

If none of ours is within 50 km when a stop ends, it moves in: its next stop is **half as far** from X36. Stay near and it keeps its distance.



IN THE LOG

"TRK 4423 doesn't want our inspector that close: it's moving to the far side of X36."

TENSION AND WAR

The escalation number at the top right is how close the crisis is to open conflict. It only goes up. At 100 you lose.

WHAT RAISES IT

+5
EACH CROWDING
Every time you push it away. Three crowdings: +15.

+12
AN HOUR INSIDE 1 KM
For every hour you stay closer than 1 km to it.

+3
IT PARKS TOO CLOSE
Each time it parks inside X36's 1 km keep-out.

WHAT HAPPENS AS IT CLIMBS



AT 25: CAUTIOUS
It gives way one time fewer, and stops creeping in when you're far away.

AT 50: IT LEAVES
The next time you crowd it, it goes home at once.

AT 100: WAR
Open conflict. The operation fails, whatever else happened.

WHAT IT COSTS YOU AT THE END

0	15	25	50	100
final escalation	three crowdings	first warning	it leaves at once	war
10 / 10 PEACE	8.5 / 10	7.5 / 10	5 / 10	YOU LOSE

THE TRADE

Pushing hard wins sooner, since high tension makes it leave early. But every point of escalation costs you score at the end, and too much is war.

The sweet spot is **between 1 and 5 km**: close enough to crowd it, far enough that the hourly +12 never starts.

WARNINGS
The event log warns you at 25, 50 and 90. The bar turns amber at 25 and red at 50.

THREE WAYS TO PLAY

There's no single right answer. These three plans all work; they trade score, speed and risk differently.

STEADY · HIGH SCORE

THE PATIENT SHADOW

Close in early and stay 2 to 4 km from Cosmos. Crowd it once at each stop. Low tension, a good track all game, a few m/s of fuel. Slow, but it scores highest.

TENSION **LOW** FUEL **MODERATE** SPEED **SLOW**

FAST · RISKY

THE HARD PRESS

Crowd it at every chance and sit near 1 km. Escalation passes 25, then 50, and it leaves at the next crowding. Quick, but you give up most of the peace points, and one slip is war.

TENSION **HIGH** FUEL **MOST** SPEED **FAST**

DEFENSIVE

THE GUARD

Park beside X36 instead of chasing Cosmos. Because you're within 50 km it keeps its distance, and your telescope holds the track. But it never leaves on its own: sooner or later you must go to it.

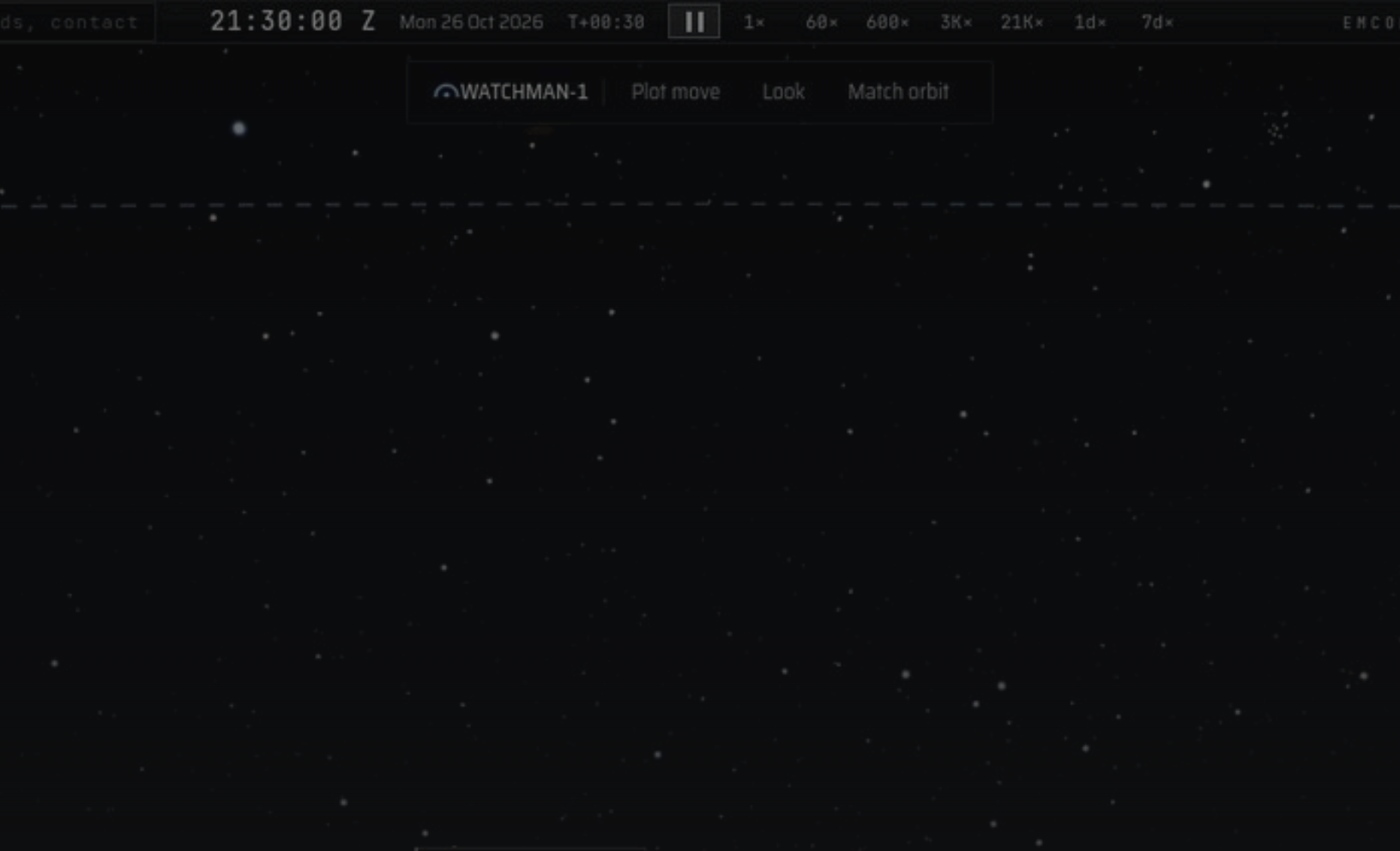
TENSION **LOWEST** FUEL **LEAST AT FIRST** SPEED **SLOWEST**

MIND THE CLOCK

Three crowdings, a stop apart, only fit into two days if you start early. Get alongside in the first few hours.

TRY IT SAFELY

BRANCH FROM NOW on the Timeline saves the moment. Try a bold move; if it goes wrong, go back (page 27).



2

YOUR FIRST GAME

The screen, the controls, and a winning game played move by move.

11 STARTING A GAME

12 THE SCREEN

13 TIME AND THE CAMERA

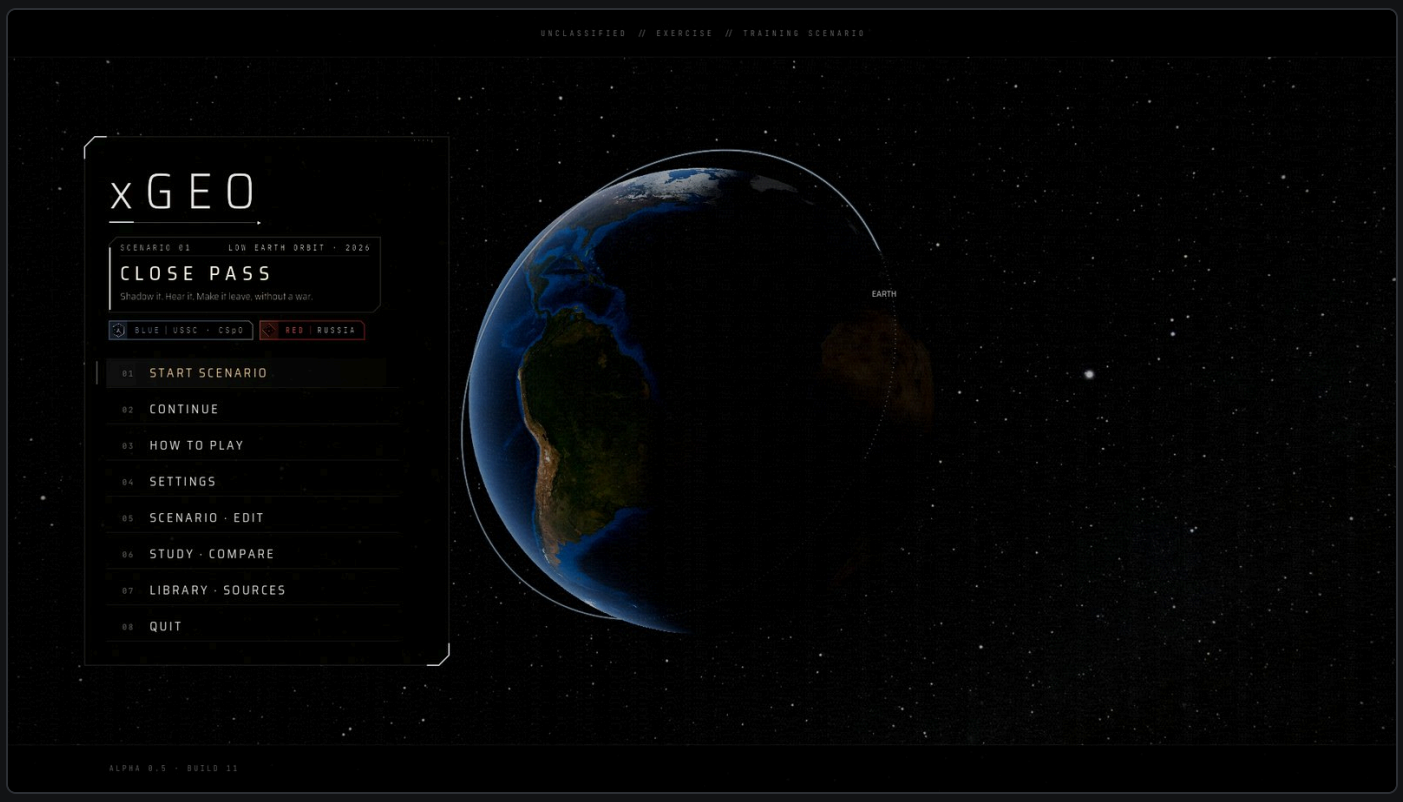
14 GET YOUR BEARINGS

15 CLOSE IN

16 SEE IT OFF

STARTING A GAME

The game opens on the live Earth turning behind the menu. Any key skips the intro.



START SCENARIO	A new game of CLOSE PASS with the tutorial on. Time starts paused.
CONTINUE	Carry on from your save. Greyed out when there isn't one.
HOW TO PLAY	A one-page briefing: goals, the four things you do, and the keys.
SETTINGS	Plain or military words, music and sound volume. Remembered between games.
EDIT · STUDY · LIBRARY	For tinkerers: change the scenario's numbers, test plans, read the sources (page 37).

SAVING

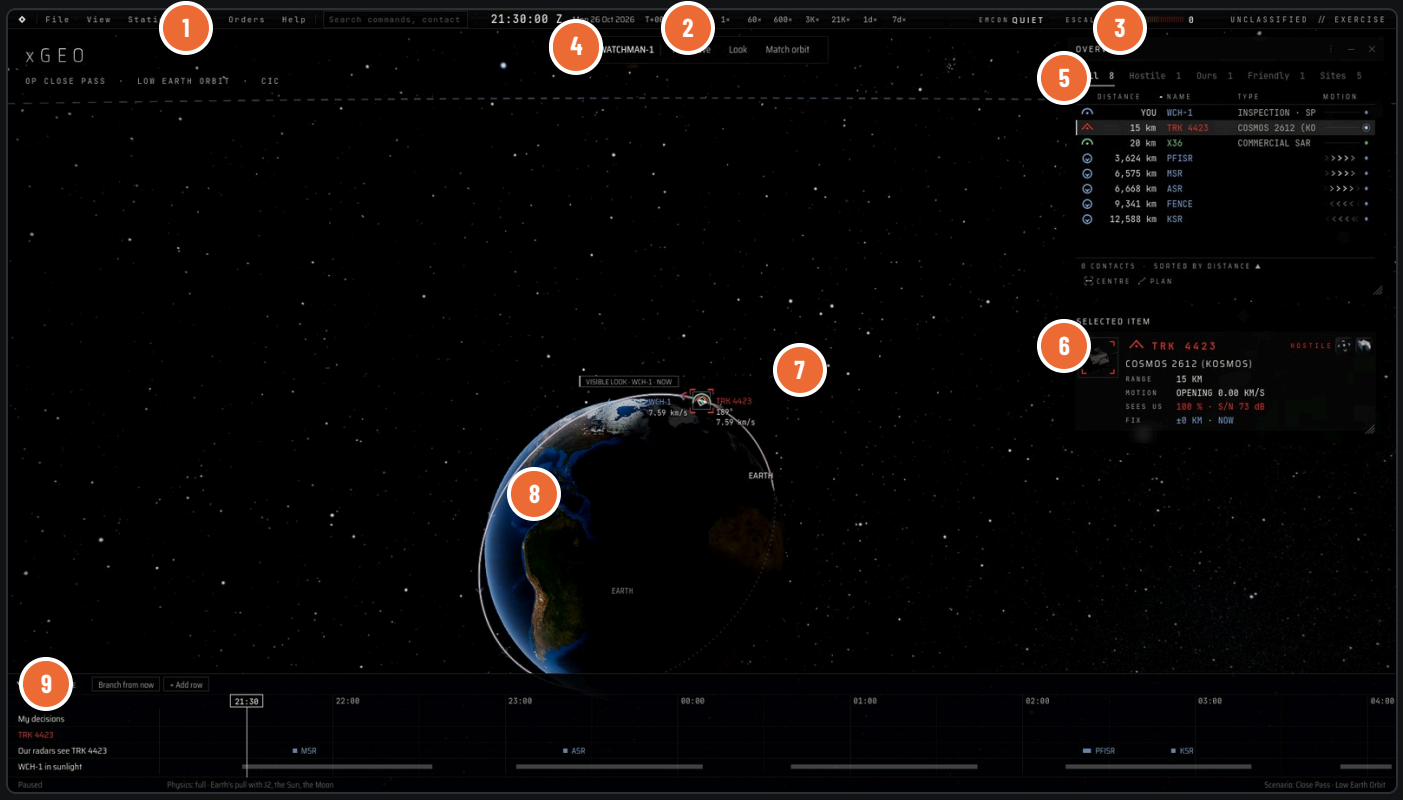
There's one save. The game writes it when you go back to the title screen (Esc with nothing open) and once every game day. Finishing a game deletes it.

PLAIN OR MILITARY WORDS

Settings › Terms switches between plain words and military terms, such as LISTENING STATION or EW CONSOLE. The game is the same either way.

THE SCREEN

Everything happens on one screen. Hover over anything and a card explains it.



1 **MENUS AND SEARCH**
Every command, with its key. Type in the box to find one.

2 **CLOCK AND TIME SPEED**
UTC time, mission time, and how fast time runs.

3 **ESCALATION**
How close the crisis is to war, 0 to 100.

4 **YOUR ORDERS**
Plot move, Look, Match orbit. Keys 1 2 3.

5 **OVERVIEW**
Every contact, nearest first. Click to select, double-click to centre.

6 **SELECTED ITEM**
Everything you know about what you clicked: range, whether it sees you, how sure you are.

7 **COSMOS 2612**
Shown as TRK 4423, your track on it: your best guess of where it is.

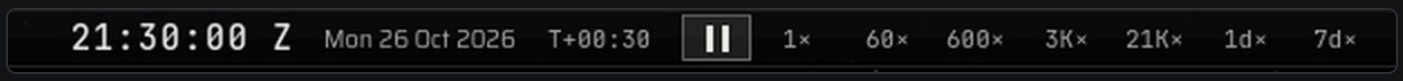
8 **THE 3-D VIEW**
The Earth, day and night, your orbit, the radars as rings on the ground.

9 **TIMELINE**
The next hours: your burns, Cosmos's closest pass, radar passes, sunlight.

2 · YOUR FIRST GAME

TIME AND THE CAMERA

The game starts paused, and time only moves when you let it. One trip round the Earth takes about 95 minutes.



Click a speed, or use [ ] to go slower or faster. Space pauses.

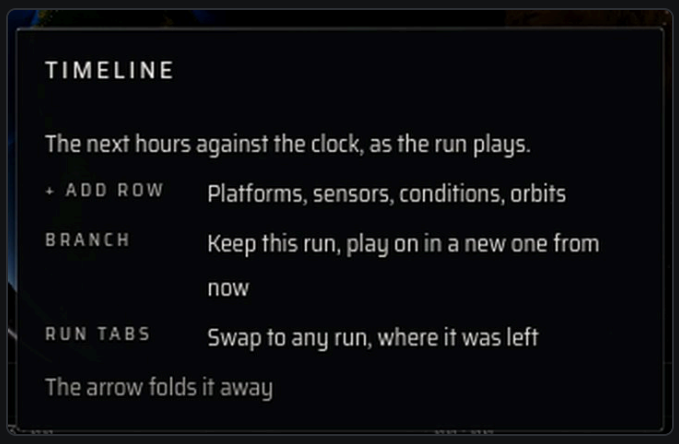
1×	60×	600×	3K×	21K×	1d× 7d×
real time	a minute a second	10 min a second	an hour a second	6 hours a second	a day, a week
WATCH A BURN	NEAR THE TARGET	COASTING	WAITING IT OUT	LONG WAITS	NOT NEEDED HERE

MOUSE

Drag	Turn the view round its focus
Right-drag	Slide the view
Wheel	Zoom in and out
Click	Select a contact
Double-click	Focus on it (ours come in close)
Right-click	Its orders

KEYS

W A S D	Turn the view
Q E	Zoom out and in
F	Close-up on the selection
Tab	Focus on the next object
Shift	Hold for faster camera keys
Esc	Back out of whatever is open



HOVER FOR HELP

Rest the mouse on any number, button or contact and a card says what it is and what it means for you. When you're unsure, hover first.

SPEED FOR THE JOB

60× near Cosmos, 600× while coasting to a meeting, 3K× to wait out a stop. Pause whenever you need to think: nothing happens while you plan.

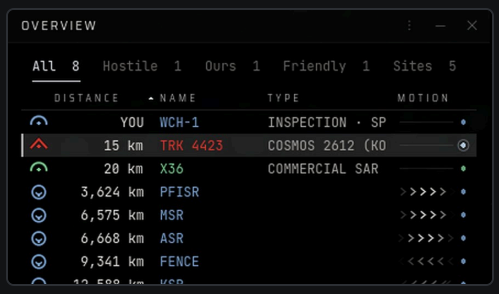
1 · GET YOUR BEARINGS

Before you move, know where everyone is, and hear Cosmos talk. Once you're within 10 km it stops.

1 MEET THE PLAYERS

Click X36, then TRK 4423, in the Overview. X36 is 20 km ahead of you. TRK 4423 is Cosmos 2612, 15 km away. Double-click each to see it.

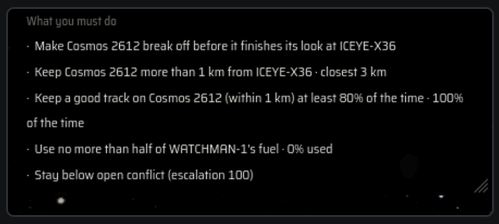
OVERVIEW · CLICK, DOUBLE-CLICK



2 READ YOUR GOALS

Five goals. Two lose the game: letting it finish, or escalation 100. The rest are points.

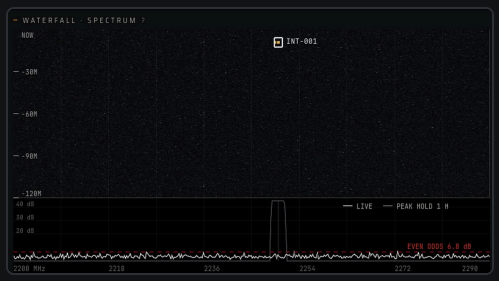
F1 CAMPAIGN · ESC CLOSES



3 LISTEN

Open the Listening station and speed up. Its first report comes within about three hours, a bright stripe at 2,250 MHz.

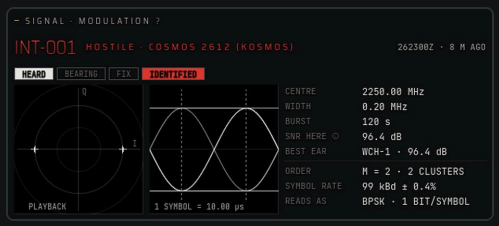
F2 ·] TO 600×



4 IDENTIFY IT

Click the recording, find its direction, identify it. Now you know who's talking, and your track got sharper. Do it within an hour.

CLICK INT-001 · D · I



5 WAIT FOR IT TO PARK

Cosmos is hopping to its first stop. About three hours in, the log says it has parked 2 km behind X36. That's your moment.

WATCH THE EVENTS, BOTTOM LEFT



2 · CLOSE IN

Get alongside it. The planner does the orbital maths; you pick how fast and how much fuel.

6 PLOT A MOVE

The planner opens on TRK 4423. Check the lamps: if the track is only known to ±10 km, a radar pass or a look will sharpen it first.

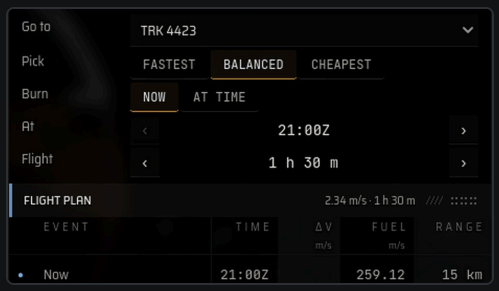
SELECT TRK 4423 · 1 PLOT MOVE



7 PICK, THEN BURN

BALANCED flies in 1½ hours for about 2.3 m/s in all. CHEAPEST waits longer for about half that. Either is a sliver of your 260 m/s.

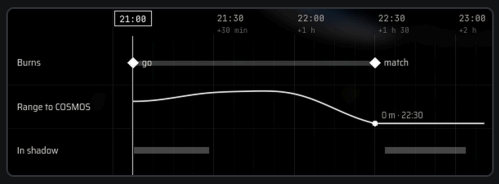
BALANCED OR CHEAPEST · BURN NOW



8 COAST

The move shows on the map as PLANNED MEETING with a countdown. Slow down for the last few minutes.

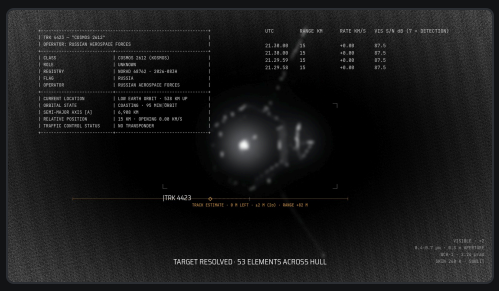
] TO 600× · 60× NEAR THE END



9 LOOK AT IT

Close in and in sunlight, the telescope shows the hull. In Earth's shadow it sees nothing: wait for the Sun.

SELECT TRK 4423 · 2 LOOK



10 MATCH ORBIT

Arriving isn't stopping. Match orbit cancels the last of your speed so you stay beside it. Stop 1 to 5 km away.

3 MATCH ORBIT, ON ARRIVAL



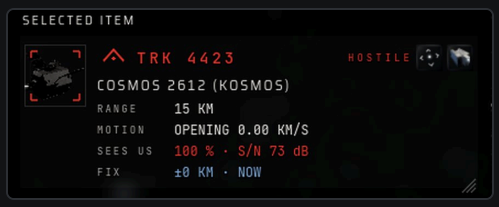
3 · SEE IT OFF

Crowd it at each stop, follow it when it moves, and the third time it gives up.

11 MAKE SURE IT SEES YOU

Crowding only counts if its telescope sees you. Check SEES US: high means it does. Then the log says it's moving to the far side of X36. Escalation +5.

SELECTED ITEM · SEES US



12 FOLLOW IT

It moves 8 km off, to the far side of X36. Once it parks, plot a move to it again and match. For a hop of a few km, a small close move is cheapest.

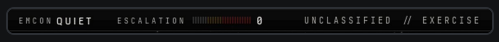
1 PLOT MOVE AGAIN · OR TACTICAL



13 MIND THE GAP

Stay more than 1 km from it. Inside 1 km, escalation climbs 12 every hour.

KEEP 1 TO 5 KM



14 CROWD IT A THIRD TIME

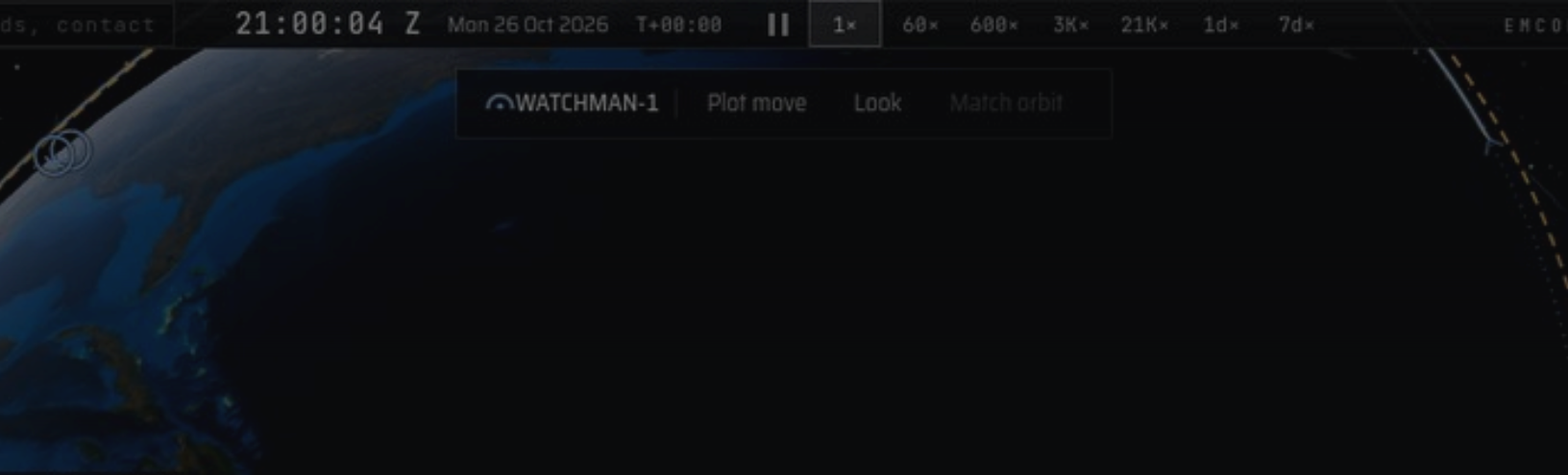
It gives up and climbs away. Once it's 200 km from X36, the operation is complete.

THEN THE DEBRIEF



IT MOVED CLOSER TO X36?
If none of ours is within 50 km when a stop ends, it takes its next stop at half the distance. Stay within 50 km.

NOTHING HAPPENED?
It only gives way while parked. Mid-hop, wait for the log to say it has parked, then close within 5 km.



3

YOUR TOOLS

One page per job: what the tool is for, how to use it, and what to watch out for.

18 KNOW WHERE IT IS

19 PLAN A MOVE

21 SMALL MOVES

22 LOOK

23 LISTEN

25 BEING SEEN

26 RADARS

27 THE TIMELINE

KNOW WHERE IT IS

You never see Cosmos 2612 directly. You see your **track** on it: a best guess, with a size, that gets less sure every minute it isn't seen.



Overview F4. Every contact, nearest first. Chevrons flow in while it closes on you, out while it opens. A beating dot means a sensor holds it now.



Selected item. What you know about the contact you clicked.

COLOURS

- Blue — Ours
- Green — Ours to protect (X36)
- Red — Hostile
- Yellow — Unknown

LOST SIGHT OF IT?

It stays where you last saw it, dashed and fading, with a chip like LAST 2H 14M. After 24 hours unseen the track is dropped.

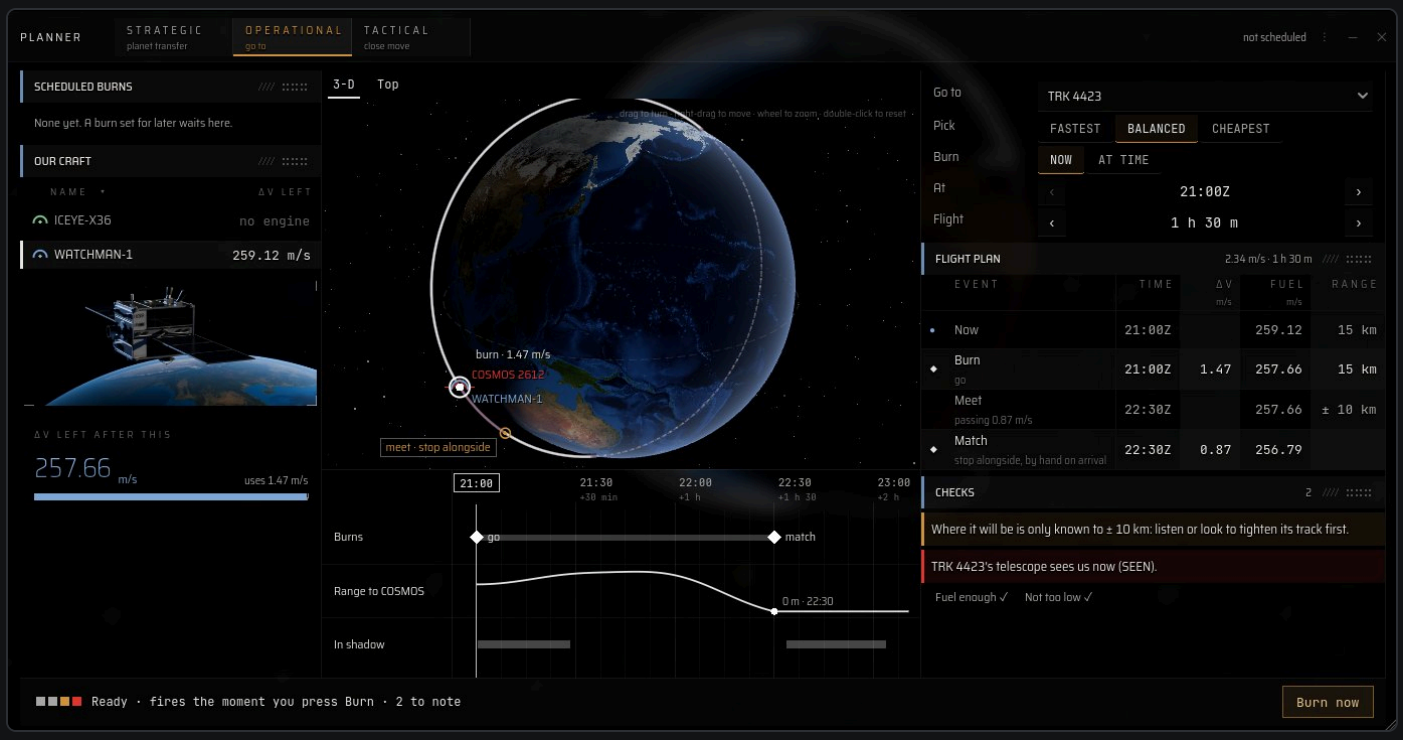
THE FOUR LINES THAT MATTER

RANGE	How far it is from you.
MOTION	Closing or opening, and how fast.
SEES US	The chance its telescope sees you on each look. **Crowding only works when this is high.**
FIX	How sure you are where it is (±), and how old the last sighting is. **Check it before you plan a move.**

> **A TRACK IS A BELIEF**
> ±10 km means you could arrive and find empty space. Listen, look or wait for a radar pass to sharpen it.

PLAN A MOVE

You say where to go. The planner works out every way there over the next day, under full gravity, and shows you the price.



Going to TRK 4423, **Balanced**: burn now for 1.47 m/s, fly 1 h 30 m, then 0.87 m/s to stop alongside. 2.34 m/s in all.

FOUR STEPS

1. **OPEN IT**
   Select a contact, press `1` or `F3`. Or right-click it › Plan intercept.
2. **CHOOSE WHERE**
   `GO TO`: a craft, an orbit, or the Moon.
3. **CHOOSE THE TRADE**
   Fastest, Balanced or Cheapest (right).
4. **BURN**
   `BURN NOW`, or set a time and it fires by itself.

TIME OR FUEL

FASTEST

Gets there soonest you can afford. Costs the most.

BALANCED · THE DEFAULT

A fair trade of time and fuel, counting the stop at the end.

CHEAPEST

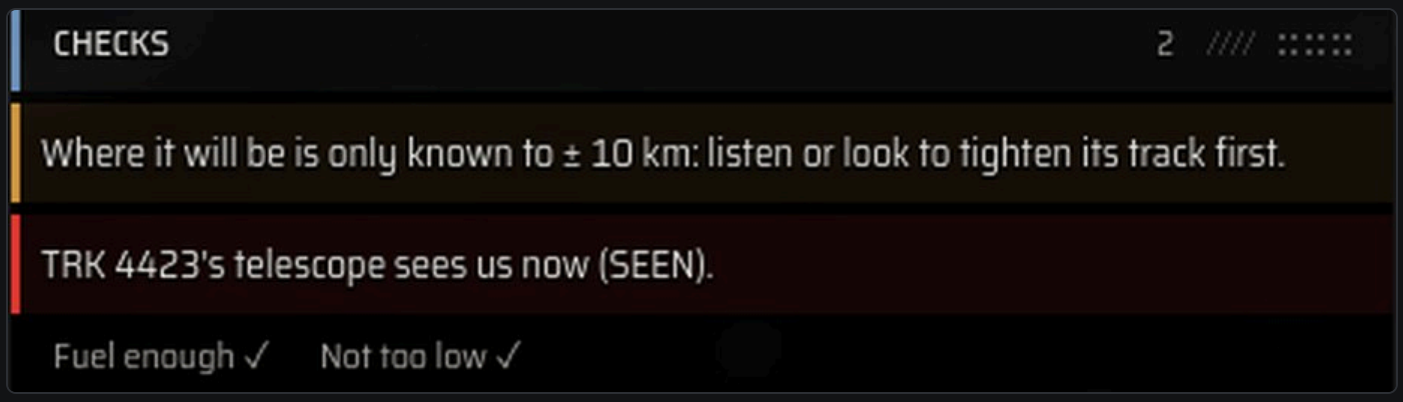
Waits and flies longer for the least fuel. Good when you have time.

ARRIVING ISN'T STOPPING

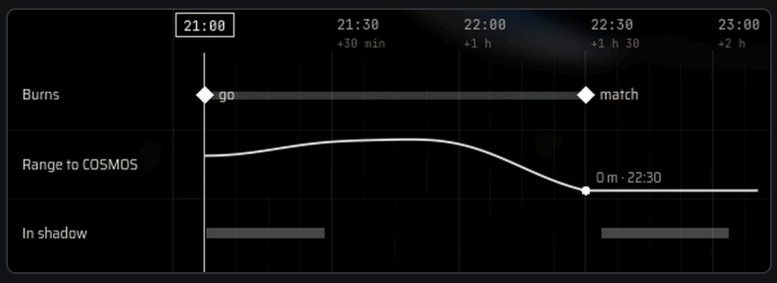
You coast past at a few tenths of a m/s. Press `3` Match orbit when you get there.

CHECK THE PLAN, THEN BURN

Before you burn, the planner checks the plan and draws it. A red lamp stops you; an amber one is a warning worth reading.



FUEL ENOUGH	You can't afford the burn at all.
FUEL TO STOP	You can get there but not stop. Plan cheaper.
NOT TOO LOW	The path dips below 122 km, into the air.
ARRIVE ± 10 KM	Your track is too vague to aim well. Sharpen it first; under 5 km passes.
SEEN · COLD, WARM, SEEN	Whether Cosmos can see you right now, and when that changes.



The plan's timeline: diamonds are burns, the range falls to zero at the meeting, and grey marks where you're in shadow.

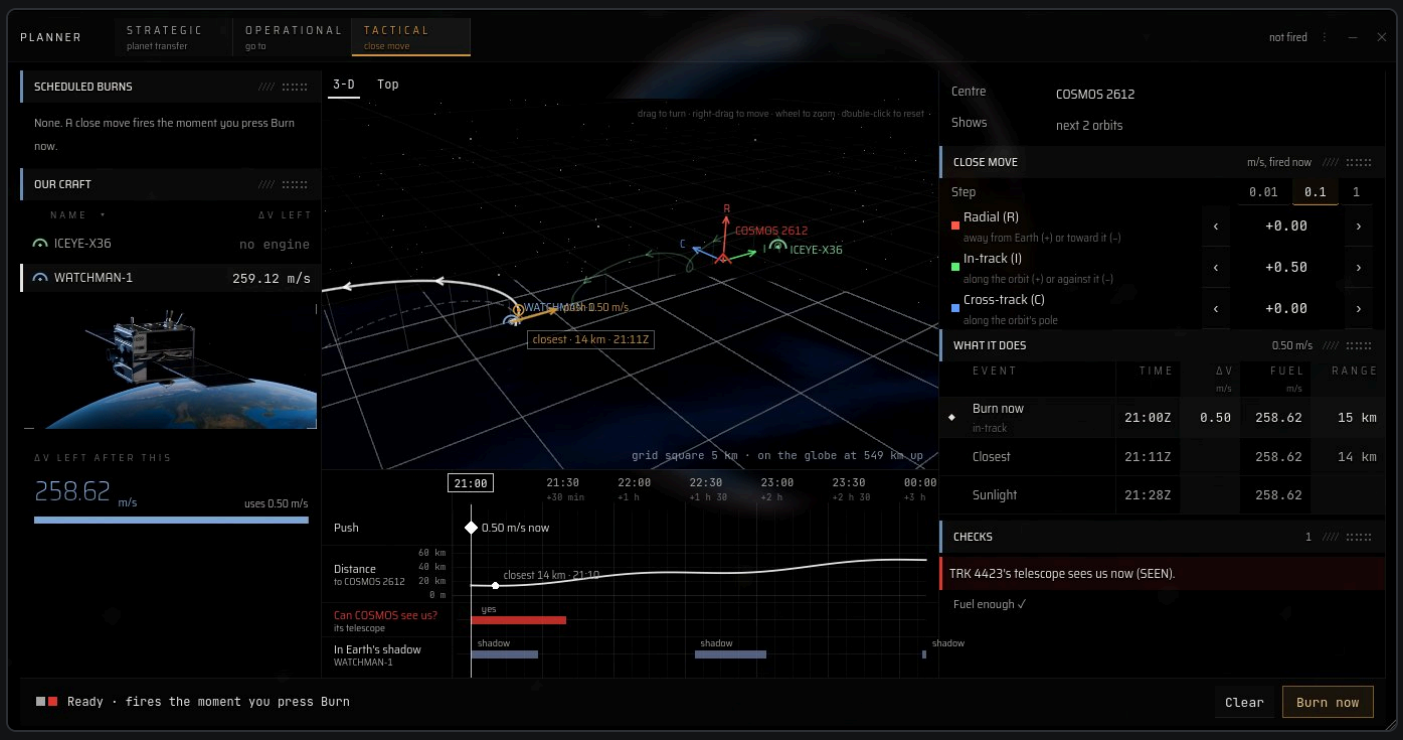
READING THE PICTURE

Soft blue	Your orbit now
White	The planned path
Dashed sand	The orbit you're going to
Dashed amber	A way you can't afford

REFUSED?
If a burn can't be flown, the log says why, for example no path at that time.

SMALL MOVES UP CLOSE

Within a few km, you don't plan transfers. You give a small push and watch where it carries you over the next two orbits.



Tactical tab: a push of +0.5 m/s along the orbit, drawn round Cosmos 2612. The amber label marks the closest pass.

ORBITS DON'T STEER LIKE CARS

ALONG THE ORBIT (I)

FORWARD MAKES YOU DRIFT BACK

Speeding up raises your orbit, and a higher orbit is slower. To drift ahead, push backward.

UP OR DOWN (R)

MAKES A LOOP

You swing in an oval round where you were, once an orbit, with no drift. Good for circling a target.

SIDEWAYS (C)

SWINGS SIDE TO SIDE

You cross back once an orbit. Rarely needed here.

OR DRAG YOUR ORBIT

Grab your own orbit line where you want to burn. Drag **along** it to speed up or slow down, **across** it to push out or in. The new orbit swings out in dashed amber with the price. Let go and the burn is set for the next time you pass that point.

Right-click or Esc cancels. Hold Shift to turn the view instead.

RULE OF THUMB

In low orbit, 0.1 m/s along the track moves you about 1.7 km per orbit. Tiny pushes, patiently, do the work.

3 · YOUR TOOLS

LOOK WITH THE TELESCOPE

WATCHMAN-1's telescope is how you see Cosmos for yourself. From far off it's a point of light. Close in, it's a hull you can name.



TRK 4423 at 15 km: **Target resolved**. Select a contact and press 2 or I; Esc closes it.

IT ONLY SEES A TARGET THAT IS

IN SUNLIGHT

In Earth's shadow there's nothing to see. The Timeline shows when it's lit.

NOT AGAINST THE EARTH

The bright Earth behind it washes it out.

CLOSE ENOUGH

It names a hull inside about 60 km.

IT WORKS WITHOUT YOU

You don't need the view open. Every 10 minutes the telescope looks at everything you track, and each sighting sharpens your track.

Close in, it also measures range from how big the hull looks: to about 30 m at 5 km.

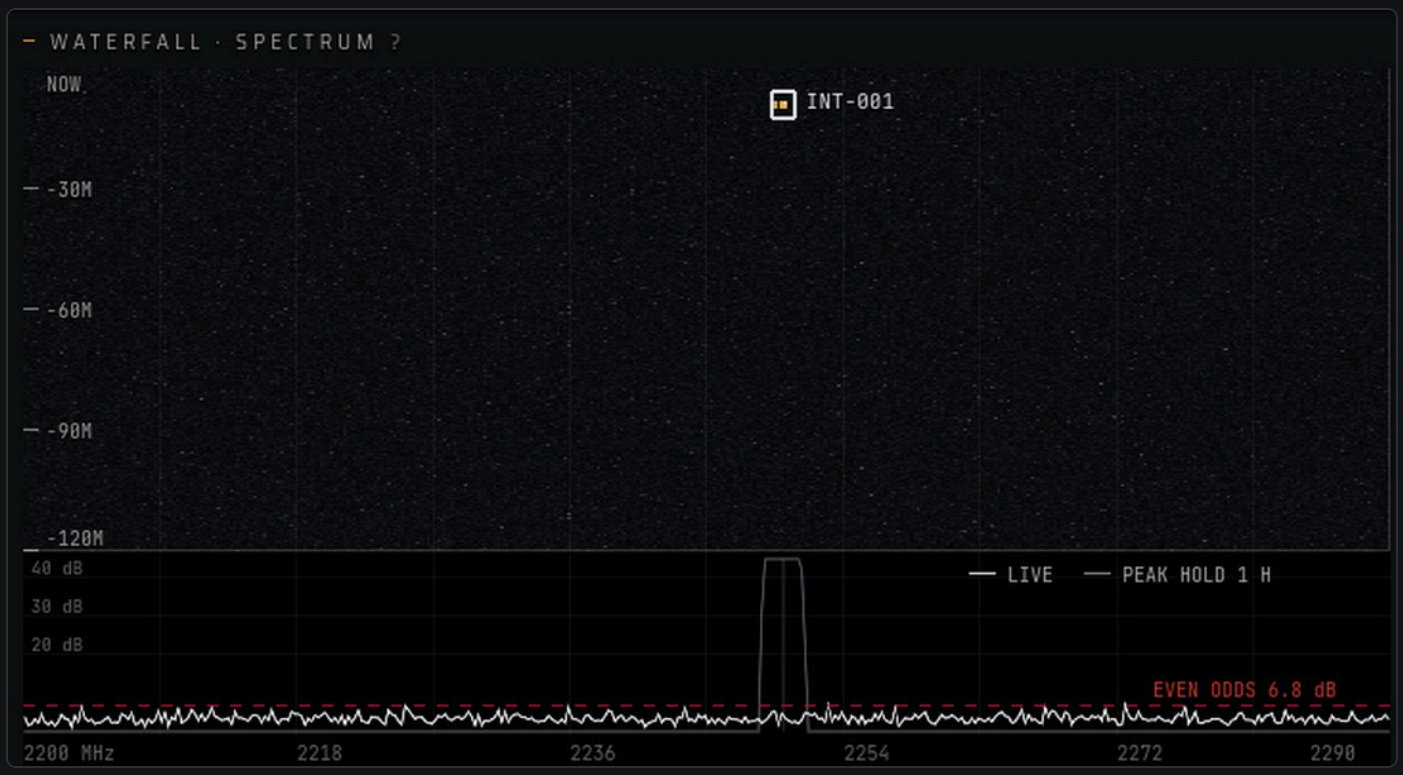
WHY STAYING NEAR PAYS

Radars see Cosmos only on their passes. In between, your telescope keeps the track good, but only on the sunlit part of each orbit.

3 · YOUR TOOLS

LISTEN TO ITS RADIO

Cosmos 2612 reports home in short radio bursts. The Listening station F2 shows each burst arriving live and records it.



The **waterfall** (time runs down, newest at the top) and the **spectrum** under it. Cosmos's burst is the peak at 2,250 MHz in S band.

WHEN IT TALKS

EVERY 3 TO 6 HOURS

And once after each move. Each burst lasts 2 minutes.

OVER RUSSIA

Only while its ground station can see it, so reports bunch together.

NOT WHEN YOU'RE CLOSE

Silent while you're within 10 km.

NOT AFTER A BURN

Silent for 6 hours after it sees one of ours burn.



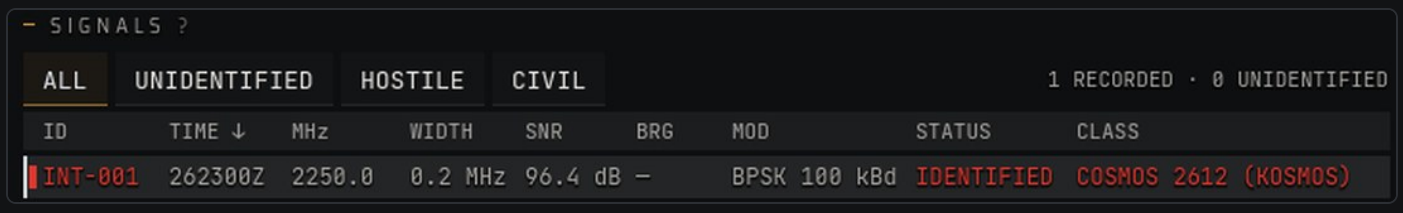
The station opens on S band, where Cosmos talks. B steps through the bands.

LISTEN FIRST

Hear and identify it while you're still 15 km out. Once you close in, it goes quiet.

IDENTIFY THE SIGNAL

A recording is a clue. Work it within an hour and you know who is talking, and where from.



The recordings list. Status goes **Heard → Bearing → Fix → Identified**.

STEP 1

CLICK THE RECORDING

Wait until the burst has ended. ↑ ↓ step through the list.

STEP 2

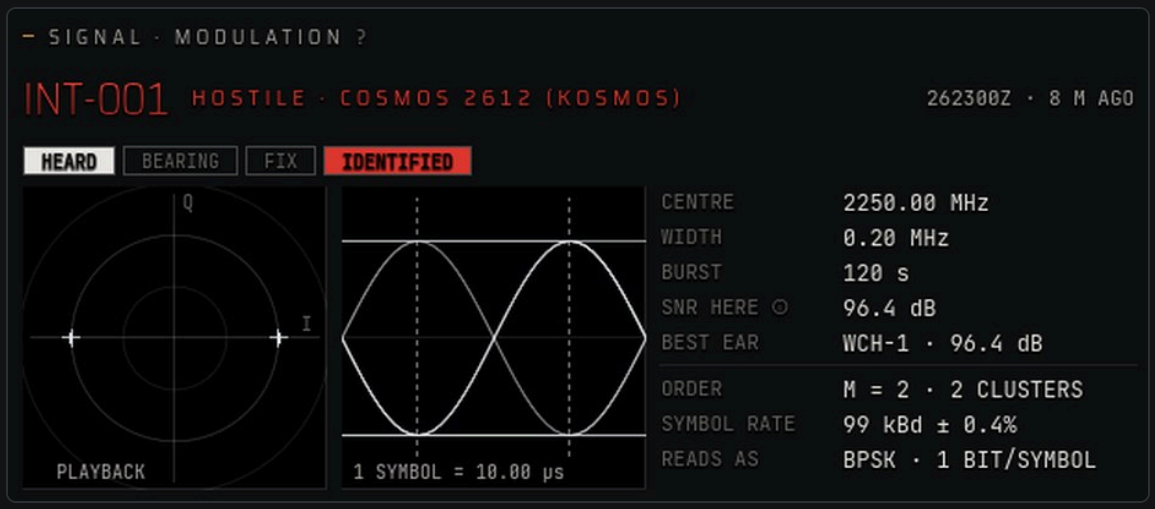
FIND DIRECTION D

A bearing from your receiver. It sharpens the track you already have.

STEP 3

IDENTIFY I

Names its class and side. Needs a clean copy: at least 15 dB.



INT-001 worked: two dot clusters read as BPSK, and the threat library matches **Cosmos 2612**.

TOO WEAK?

The hint says so. Wait for a stronger report, or get closer, but not within 10 km.

SHORTCUT

ANALYSE AUTOMATICALLY works each recording as it arrives. Handy, but you learn nothing.

BEING SEEN

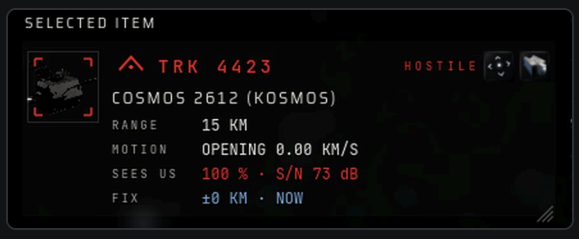
Cosmos watches you with a small infrared telescope. Here, being seen is mostly good: crowding only works if it can see you.



Thermal forecast F7: your orbit over the next pass, coloured by how plainly Cosmos sees you.

COLD	Hidden. It almost never sees you.
WARM	Seen now and then.
SEEN	Seen on most looks.

The quickest check is **SEES US** in the Selected item. If it reads 0 %, crowding won't count: wait for a better angle.



Select TRK 4423 to see **SEES US**.

READING THE FORECAST

YOUR HEAT NOW	Hidden, Detected or Seen, with the range it sees you best from.
IN SHADOW	Whether you're in Earth's shadow now, and when the Sun comes back.
QUIET BURN	When a burn would go unseen. "None" means any burn this orbit shows.
HIDDEN FOR	How long you've been out of sight.

SEEN = GOOD, WHEN CROWDING
Park within 5 km while it sees you, and it gives way.

BURNS ARE SEEN
If it sees you burn, it goes quiet on the radio for 6 hours. Listen first, then move.

SHADOW HIDES YOU
Arriving in Earth's shadow may not count until the Sun comes round.

3 · YOUR TOOLS

RADARS AND YOUR TRACK

Five real ground radars fix Cosmos to within metres, but only while it passes overhead. Between passes, your track drifts.

PFISR · ALASKA	Poker Flat. Looks high: from 45° up, out to 3,000 km.
MSR · TEXAS	LeoLabs Midland. From 30° up, out to 3,000 km.
ASR · ARIZONA	LeoLabs Arizona. From 20° up, out to 3,000 km.
KSR · NEW ZEALAND	LeoLabs Kiwi. From 40° up, out to 1,500 km.
FENCE · KWAJALEIN	The Space Fence. From 20° up, out to 3,000 km.



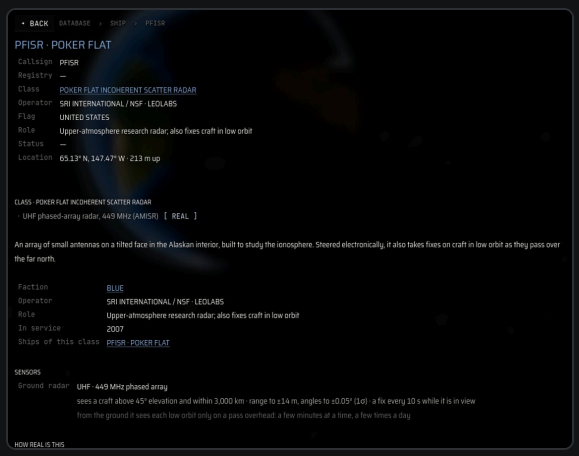
The Timeline row **OUR RADARS SEE TRK 4423** shows each pass. A pass lasts a few minutes; there can be hours between them.

HOW A TRACK LIVES

SHARPENS
EACH SIGHTING
Radar, telescope or radio bearing: each one tightens it.

WIDENS
BETWEEN SIGHTINGS
Small unknowns grow. Unseen burns widen it more.

DROPPED
AFTER 24 HOURS
Unseen that long, or too vague to say where it is.



EACH RADAR HAS A PAGE

Open the Database F5 and pick a radar to see where it is, how high it looks, how far it reaches, and the source for each number.

PLAN ROUND THE PASSES
Arrive just after a radar pass and your track is at its sharpest. Arrive after a long gap and check the ± first.

THE TIMELINE

Along the bottom: the next hours, on the same clock as everything else. It tells you when to act.



The boxed time is now. `BRANCH FROM NOW` and `+ ADD ROW` sit on its title line.

MY DECISIONS	Burns you've set, and the move the planner has open.
TRK 4423	Its closest pass to anything, such as `CLOSEST · 2 KM · X36`.
OUR RADARS SEE TRK 4423	Each radar pass.
WCH-1 IN SUNLIGHT	When you're lit. The gaps are Earth's shadow.

ADD A ROW

`+ ADD ROW` adds up to nine: distances, heights, sunlight, when each radar sees each craft, when you're within 1, 5 or 10 km of Cosmos.

> **WORTH ADDING**
> WCH-1 within 5 km of TRK 4423. It shows exactly when you're crowding.



BRANCH FROM NOW

Saves this moment as Run 1 and plays on as Run 2. Click a run's tab to go back exactly where you left it. Branch before a big move. Branches aren't saved to disk.

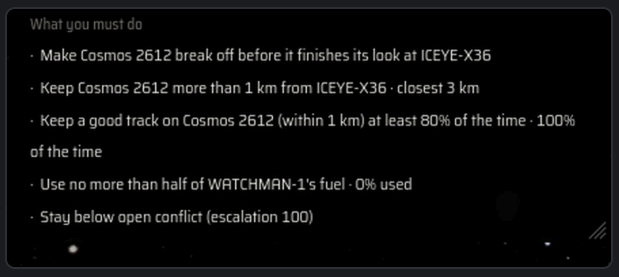
EVENTS, BOTTOM LEFT

Warning: act now. **Caution**: worth knowing, like losing sight. **Nominal**: good news, it gave way. **Advisory**: for the record.

3 · YOUR TOOLS

OTHER WINDOWS

You can win without these, but they answer the questions the main screen doesn't.



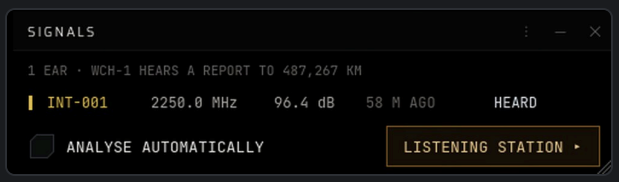
CAMPAIGN F1

Your five goals, each ticked, crossed or open, with progress so far.



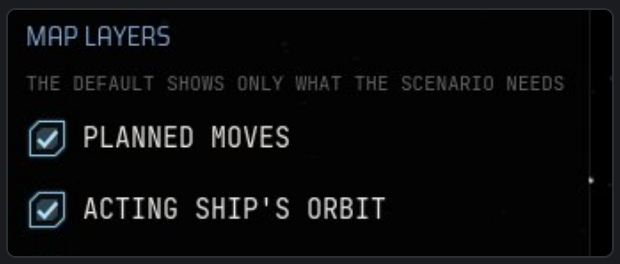
DATABASE F5

A page on every craft and radar: what it carries, how much fuel, how far it sees, and where the numbers come from.



SIGNALS

A small window with the newest recordings. Click one to open the Listening station on it.



MAP LAYERS L

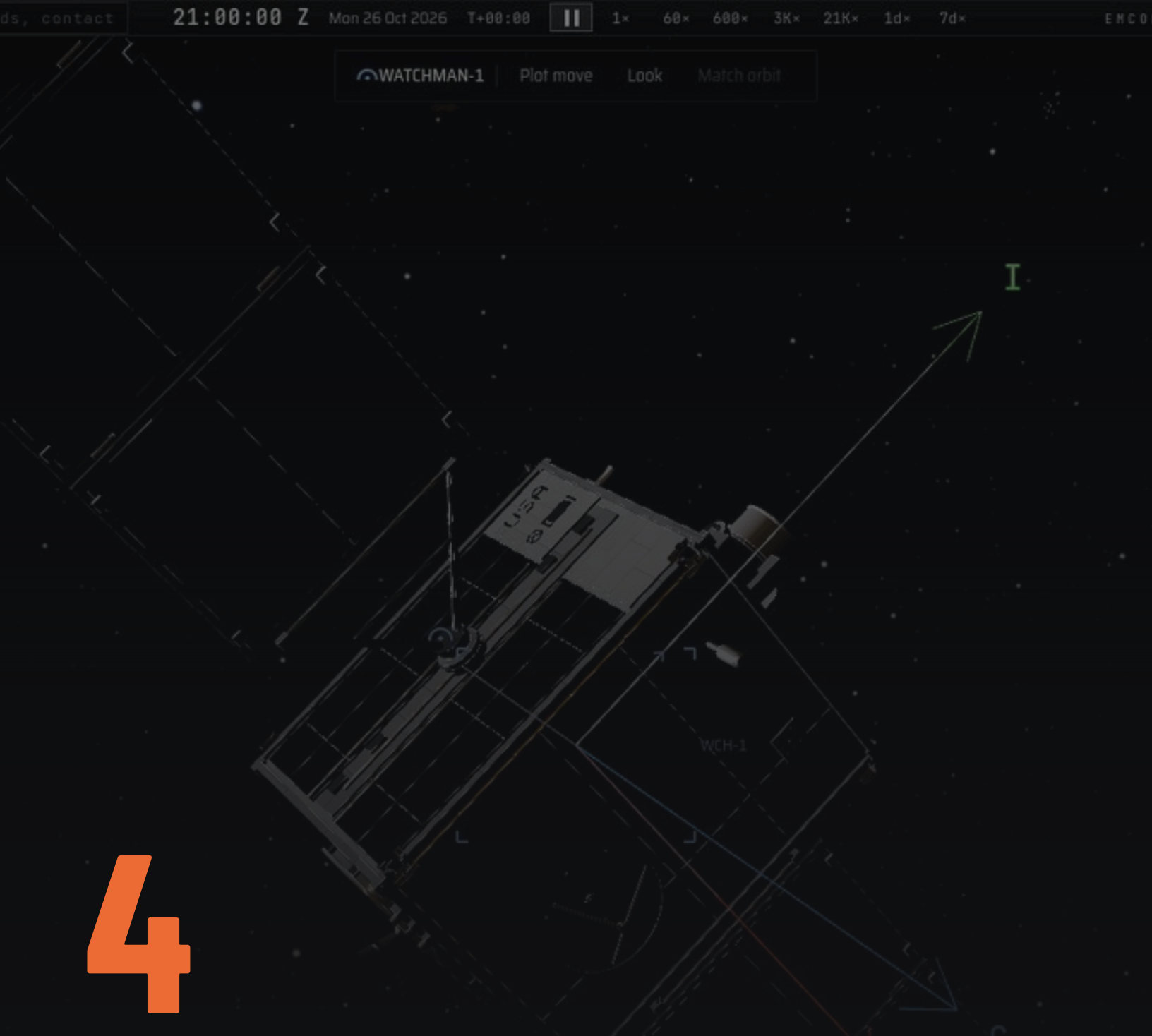
Tick what the 3-D view draws: planned moves, orbits, detections, the close-up frame.

SOLAR SYSTEM

Stations › Solar system. A live chart of the Sun and planets, with each one's distance and the cheapest trip there.

THE SUMS

Click SNR HERE ⓘ in the Listening station to see the radio maths, term by term. Military terms add formulas to hover cards.



4

REFERENCE

Tips, answers, the keys on one page, and every word explained.

30 PLAYING WELL

31 THE DEBRIEF

32 PHYSICS YOU'LL FEEL

33 PROBLEMS AND ANSWERS

34 CHEAT SHEET

35 GLOSSARY

PLAYING WELL

Twelve things good players learn, most of them the hard way.

01
LISTEN FIRST
It stops talking once you're within 10 km or it sees you burn. Identify it from 15 km out.

02
MOVE WHEN IT PARKS
A target mid-hop is a moving goal. Wait for "has parked", then plan.

03
ALWAYS MATCH
A transfer passes by. Forget Match orbit and you sail past at a metre a second.

04
USE THE SUNLIGHT
Your telescope only sees a lit target. The Timeline's sunlight row tells you when.

05
BE SEEN WHEN CROWDING
Check Sees us. Arriving in shadow may not count until the Sun comes round.

06
CHECK THE FIX
±10 km means you may arrive somewhere empty. Sharpen the track first.

07
CHEAP AND EARLY
Small moves early cost less than big ones late. Let the orbit do the work.

08
1 TO 5 KM
The sweet spot: it gives way, and tension rises only 5 a time.

09
BRANCH BEFORE BIG MOVES
Try it; if it goes wrong, go back to Run 1.

10
STAY WITHIN 50 KM
Out of reach, it creeps toward X36 and can enter the keep-out.

11
PICK THE RIGHT SPEED
60× near it, 600× coasting, 3K× to wait out a stop. Pause to think.

12
MIND THE CLOCK
Three crowdings, a stop apart, fit in two days only if you start early.

4 · REFERENCE

THE DEBRIEF

When the operation ends, the view holds on the deciding moment, then the debrief shows what happened and how you scored.



WHAT IT SHOWS

THE RESULT	And the one line that decided it.
SCORE / 100	Each goal's points, final escalation, how long it took.
WHAT HAPPENED	A replay, and every event about Cosmos. Click a line to jump there.
ADVICE	One tip for each goal you missed.

THE REPLAY

X36 sits in the middle with its 1 km keep-out ring. Your path is blue, Cosmos's red, and the closest moment is ringed.

PLAY	The whole mission in 20 seconds
CLOSEST	Jump to the closest moment
Wheel **Drag**	Zoom and move

BACK TO THE TITLE

Leaving the debrief ends the game and deletes the save.

PHYSICS YOU'LL FEEL

Everything in xGEO is computed from real physics. You don't need the equations, but these six ideas explain what you see.

FALLING ROUND THE EARTH

An orbit is a fall that keeps missing the ground. At 540 km you move at 7.6 km/s and go round in 95 minutes. Everything near you falls the same way, so up close things barely move relative to each other.

FASTER MEANS SLOWER

Push forward and you climb on the far side of the Earth. A higher orbit takes longer, so you end up behind. To catch something ahead, slow down first.

FUEL IS SPEED

Fuel is counted as the change in speed it can buy: Δv. WATCHMAN-1 has about 260 m/s. A move to a craft 15 km away costs 2 or 3 m/s. Going to a geostationary orbit would cost thousands.

LIGHT AND SHADOW

Up to a third of each orbit is in the Earth's shadow. A telescope sees by reflected sunlight, so in shadow a satellite vanishes. Infrared sees heat, which fades more slowly.

RADIO FADES WITH DISTANCE

Twice as far, a quarter as strong. Whether you hear a burst depends on power, antennas, distance and noise. Identifying it needs a clean copy, 15 dB above the noise.

KNOWING ISN'T SEEING

Your track is a best estimate with an honest size. Lose sight and the ring grows; a radar pass or a look shrinks it. The Earth's bulge and the Sun and Moon all pull, so paths aren't perfect ellipses.

PROBLEMS AND ANSWERS

The usual snags, and what to do.

THE F KEYS DO SOMETHING ELSE

Click inside the game once. The same windows are on the Stations menu.

MATCH ORBIT IS GREY

Select a contact and get within 5,000 km. Hover over it: the card says what's missing.

I CAN'T HEAR ANYTHING

Reports come every 3 to 6 hours, only over Russia, and never while you're within 10 km. Run at 600× and watch the waterfall.

IDENTIFY WON'T WORK

Wait for the burst to end. It needs 15 dB and must be less than an hour old.

THE TELESCOPE SEES NOTHING

The target is in shadow, or the Earth is behind it. Wait for the sunlight row on the Timeline.

I CROWDED IT AND NOTHING HAPPENED

It must be parked, you within 5 km, and it must see you (Sees us).

THE PLANNER SAYS IT CAN'T

Nothing in the next day is affordable, or every path dips too low. Try another target, or wait.

THE GAME IS SLOW

Close windows you don't need, turn off Ship shadows, use a smaller browser window.

WHERE'S MY SAVE?

Continue on the title screen. It lives in your browser's storage for this site.

FOUND A BUG?

Use Report a bug on the xGEO site: what you did, what you expected, what happened, and the mission time.

ESC, IN ORDER

Closes a menu; cancels an orbit drag; leaves a close-up; closes the telescope; closes the newest window; and with nothing open, saves and goes to the title screen.

CHEAT SHEET

Print this page and keep it by the keyboard.

A WINNING ORDER

1. Click X36, then TRK 4423 in the Overview
2. F2 and] to 600×: wait for its report
3. Click the recording, D, I
4. Wait for "has parked"
5. Select TRK 4423, 1, Balanced, Burn now
6. 3 Match orbit 1 to 5 km from it
7. It gives way: plot and match again
8. Third crowding: it leaves

TENSION

+5	Each crowding
+12 / h	You inside 1 km of it
+3	It parks inside 1 km of X36
25 · 50 · 100	Cautious · leaves · war

SCORE

40 it leaves · 20 off X36 · 20 good track · 10 fuel · 10 no war

KEYS

Space	Pause / resume
[ ]	Slower / faster
1 2 3	Plot move · Look · Match orbit
F1	Campaign
F2	Listening station
F3	Planner
F4	Overview
F5	Database
F7	Thermal forecast
L	Map layers
I	Telescope
F	Close-up on the selection
Tab	Next object
W A S D Q E	Turn and zoom the view
Esc	Back, close, or the menu

IN THE LISTENING STATION

↑ ↓	Pick a recording
D	Find direction
I	Identify (needs 15 dB)
B	Next band

MOUSE

Drag	Turn · right-drag slides · wheel zooms
Double-click	Focus on it
Right-click	Its orders
Drag orbit	Plan a burn by hand

GLOSSARY

Settings › Terms switches between plain words and military ones. Both are here.

2Σ (TWO SIGMA)
A size that holds the truth most of the time. Track rings and Fix use it.

APOGEE · PERIGEE
The highest and lowest points of an orbit.

BEARING
The direction a signal came from.

BPSK · QPSK
Ways of sending bits by shifting a radio wave. Count the dot clusters to tell them apart.

CONTACT · TRACK
Something you know is there, and your estimate of where it is.

CROWDING
Parking within 5 km of Cosmos while it is parked and sees you.

CUSTODY
Keeping a good track: sure, and right, to within 1 km.

DB
A ratio on a log scale. +3 dB is double, +10 dB ten times.

DF
Direction finding: taking a bearing on a signal.

ΔV (DELTA-V)
The change in speed you can still make: your fuel, in m/s.

EMCON
Emission control. Quiet: you only listen. Loud: you transmit.

ESCALATION
How close the crisis is to open conflict. 100 ends it.

EW CONSOLE
The military name for the Listening station.

IN-TRACK · RADIAL · CROSS-TRACK
Along the orbit · up and down · sideways. I, R and C.

KEEP-OUT
The space kept clear round a craft. X36's is 1 km; Cosmos's is 5 km.

LINK BUDGET
The sum of gains and losses from a transmitter to a receiver.

MATCH ORBIT
The burn that cancels your speed relative to a target, so you stay beside it.

NORAD NUMBER
A satellite's number in the public catalogue. Cosmos 2612 is 68762.

RPO
Rendezvous and proximity operations: one craft working close to another.

SAR
Synthetic aperture radar: how X36 images the ground through cloud.

SIGINT
Signals intelligence: listening to other people's radio.

SNR
Signal-to-noise ratio: how far a signal stands above the hiss.

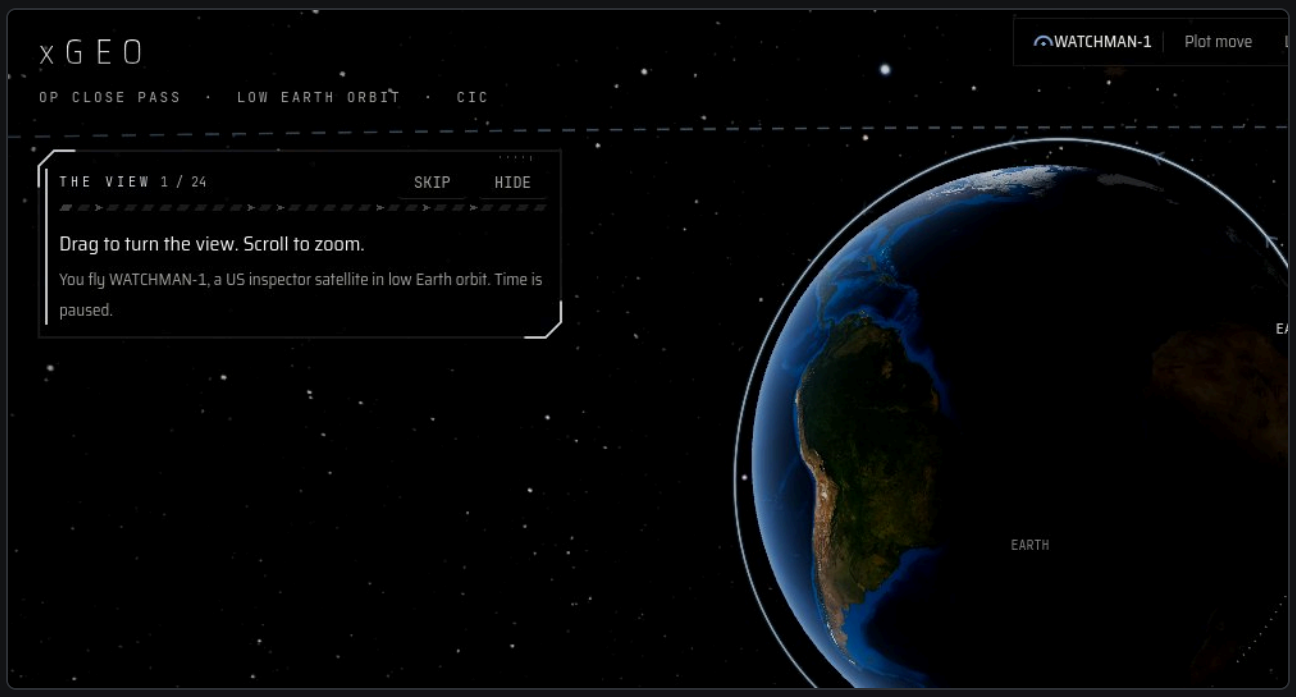
SPECTRUM · WATERFALL
Signal strength by frequency now; and the same over time.

TT&C
Telemetry, tracking and command: a satellite's radio link home.

UTC · Z
The clock satellites run on. 21:00Z is 21:00 UTC.

THE TUTORIAL AT A GLANCE

A new game starts with the tutorial card at the top left. Each step ends when you've done it. In the tutorial Cosmos leaves at the second crowding, so a full run is shorter.



SKIP moves on. **HIDE** tucks the card away.

1–2	Turn and slide the view
3–6	Find X36 and TRK 4423; the Ours tab
7–10	Database, Campaign, Map layers, Terms
11	Let time run at 60×
12–16	Listen, hear it, identify it, the link budget, Signals
17–18	Plot a move and burn
19–20	Hover over it; look through the telescope
21	Match orbit within 5 km
22	Right-click it: every option
23	Follow it and match again
24	Let time run to the debrief

FOR TINKERERS

xGEO is a simulation first. Three pages on the title screen let you change the numbers and test your plans.

EDIT

Change the scenario's main numbers: where everyone starts, WATCHMAN-1's fuel and telescope, Cosmos's habits, the tension rules, the score weights. Press **PLAY EDITED**.

STUDY

Play each plan many times (Do nothing, Shadow, Guard) and compare how often each wins. Each run redraws Cosmos's habits, so luck shows up. Desktop program only.

LIBRARY

Every source the numbers rest on, with why it was chosen. In game it's under Help.

WHAT EDIT CAN CHANGE

START	Where WATCHMAN-1 and Cosmos start, the catalogue error, the operation's length.
WATCHMAN-1	Propellant, exhaust speed, telescope size.
COSMOS 2612'S BEHAVIOUR	How close counts as crowding, how far it evades, how many times, when it feels unwatched, its closest approach.
RULES AND TENSION	The keep-out, how good a track must be, tension per crowding and per hour, when it leaves, when it's war.
SCORE WEIGHTS	The points for each of the five goals.

REMEMBER

Start scenario always plays the scenario as published. Only Play edited uses your changes. A study tells you which plan does better in this model, not what would happen in real life.

REAL AND INVENTED

ICEYE-X36 and Cosmos 2612 are real satellites, flown from their public orbits and published specifications. The five ground radars are real sites. WATCHMAN-1 is invented, loosely modelled on the US GSSAP inspectors, and what happens in the game is fiction. Their operators have not reviewed or endorsed this game.

xGEO Core, the engine under the game, is open source. Its Research Edition describes every model and the checks that hold it to its sources. Both are on the xGEO site.

xGEO Live Operator's Handbook · Alpha 0.5, build 11 · November 2026 · © 2026 Kevin Williamson. All rights reserved.